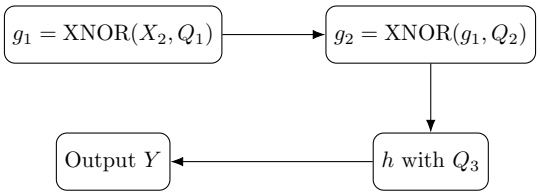
VLSI Testing: 2014 Final

Reconstructed questions and independently derived solutions. Personal names, identifiers, handwritten answers and administrative headers are omitted. Figures are redrawn as logical diagrams or explicit netlists. A source-reading note identifies a meaningful ambiguity instead of silently changing the question. Finite Boolean checks are supplied in the examination collection's Python verification script.

1 Full-scan conversion and stuck-at tests

Question. Convert the sequential circuit to full scan, derive tests covering all stuck-at faults of the combinational logic, and describe the application sequence. The reconstructed normal-mode netlist is:

Signal	Defining expression
g_1	$\overline{X_2 \oplus Q_1}$
g_2	$\overline{g_1 \oplus Q_2}$
h	$g_2 \oplus Q_3$
Y	$\overline{X_1 + h + Q_4}$
D_1	g_2
D_2	Q_1
D_3	g_1
D_4	Q_3



Solution. Replace every D flip-flop by a muxed scan flip-flop: $D_i^{\text{scan}} = SE?SI_i : D_i$. Connect $SI \to FF_1 \to FF_2 \to \cdots \to FF_n \to SO$ with common scan enable and a controllable clock. Scan state outputs are independent pseudo-inputs; all functional D inputs become observable pseudo-outputs.

$(X_1, X_2, Q_1, \ldots, Q_n)$	$(D_1, \ldots, D_n)$	Y
000000	0010	1
001011	1101	0
010110	0001	0
100000	0010	0
000100	1010	0

Exhaustive enumeration of all 2^6 input assignments verifies 54/54 single stuck-at faults for this netlist: both values on every signal stem and on every individual gate-input branch. Identity buffers at the observed D pins also model their separate output branches. No modeled faults are redundant. The displayed set is sufficient, not claimed minimum.

For each row: assert $SE = 1$; shift $Q_n, \ldots, Q_1$ in that order; set X_1, X_2; deassert SE and observe Y before the capture edge; apply one functional capture clock; reassert SE and unload $D_n, \ldots, D_1$ while loading the next state. Account for scan-in/out bit order and compare only the expected response positions. The proof covers combinational SAFs, not all scan-mux, clock, reset or analog defects.

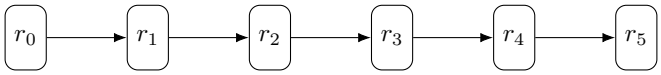
Clock skew extension. The source additionally gives 0.5 ns data-path delay from register 1 to 2 and a capture clock at register 2 delayed by 1 ns. For a same-edge transfer, the hold constraint is $t_{cq,\min} + t_{cd} \geq t_{\text{skew}} + t_{\text{hold}}$. If the given 0.5 ns is the entire minimum launch-to-data delay, it is shorter than the positive 1 ns skew even before adding hold time: a hold race can occur. Insert sufficient data-path delay, reduce/repair clock skew, or use an appropriate lockup latch for scan shifting across clock domains. Increasing the clock period alone does not fix this hold violation. If 0.5 ns describes only interconnect, $t_{cq,\min}$ and hold time are still needed for a definitive numerical margin.

2 LFSR quotient stream and signature

Question. Compute the quotient and final signature for two six-stage LFSRs, feeding the rightmost input bit first, for

$$M = \texttt{101101110110010010001010}.$$

The first circuit distributes the last-stage feedback into XOR taps; the second XORs selected register outputs into the serial input.



Solution. The drawing does not state a seed, so assume a cleared register. Number stages $r_0, \ldots, r_5$ from input to output and read the outgoing r_5 *before* each shift. No extra flush clocks are applied. All sums below are XORs over GF(2).

For the distributed-feedback circuit, let $f = r_5$. First form $(r'_0, \ldots, r'_5) = (u, r_0, r_1, r_2, r_3, r_4)$, then XOR f into positions $\{0, 2, 4, 5\}$. For the input-feedback circuit,

$$r'_0 = u + r_0 + r_1 + r_3 + r_5, \qquad r'_i = r_{i-1} \quad (1 \leq i \leq 5).$$

These recurrences, rather than a visually reversed polynomial label, define the bit ordering unambiguously.

Circuit	Serial output, first bit first	Final $(r_0, \ldots, r_5)$
Distributed	`000000111001001100000111`	`011001`
Input feedback	`000000111001001100000111`	`100011`

The first six outgoing bits are the initial zero state. Deleting that six-bit latency gives the emitted quotient portion during the remaining input clocks. The table gives physical register signatures; Fibonacci and distributed-feedback state encodings differ although their outgoing streams agree. Reversing the signature, clocking six extra zeros or using a nonzero seed changes the printed result and must be stated.

3 Validity of March sequences

Question. Which sequences have consistent read expectations on a fault-free memory?

$$a : \{\Uparrow(w0); \Uparrow(r0, w1); \Downarrow(r0, w0, r0)\},$$
$$b : \{\Uparrow(w0); \Uparrow(r1, w1); \Downarrow(r1, w0, r0)\},$$
$$c : \{\Uparrow(w1); \Uparrow(r1, w1); \Downarrow(r1, w0, r0)\},$$
$$d : \{\Uparrow(w1); \Uparrow(r1, w0); \Downarrow(r0, w1, r1)\}.$$

Solution. Only c, d are consistent. In a, the second element leaves 1 in every cell, contradicting the next $r0$. In b, the first $r1$ follows an initialization to 0. In c, each descending visit reads 1, writes 0 and reads 0. In d, the ascending pass leaves 0; the descending pass reads 0, writes 1, then reads 1. Consistency does not itself imply completeness for any fault model.

4 Equal SAF and transition-fault coverage

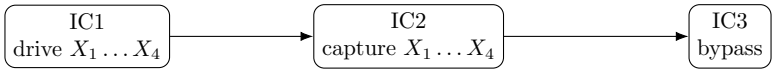
Question. Compare

$$a : \{\Uparrow(w0); \Uparrow(r0, w1); \Downarrow(r1, w0)\},$$
$$b : \{\Uparrow(w1); \Uparrow(r1, w0); \Downarrow(r0, w1, r1)\},$$
$$c : \{\Uparrow(w0); \Uparrow(r0, w1); \Downarrow(r1, w0, r0)\}.$$

Solution. b and c both detect SA0, SA1, failed $0 \to 1$ transitions and failed $1 \to 0$ transitions. They are data-complement versions of one another. In a, $0 \to 1$ is read in the last element, but the final $1 \to 0$ is not followed by a read, so a failed falling transition can escape. The initialization and first read in each case also expose both stuck values.

5 IEEE 1149.1 interconnect testing

Question. Three ICs form the serial chain $TDI \to IC1 \to IC2 \to IC3 \to TDO$. The four lines $X_1, \ldots, X_4$ run from IC1 outputs to IC2 inputs. The course-specific instruction encodings are Bypass=000, InTest=001 and ExTest=010. The question requests driving 0000 to detect SA0 on $X_1, \ldots, X_4$.



Solution. Put IC1 and IC2 in EXTEST; put the uninvolved IC3 in BYPASS. Preload safe drive values, update the launch cells, allow interconnect settling, capture at IC2, then shift out the captured response. Drive 1111 to detect SA0; drive 0000 to detect SA1. Other boundary cells must have safe values, especially output enables, to prevent contention.

Correction to the printed request. A line driven to 0 has the same value under SA0, so 0000 cannot detect any of the four SA0 faults. Use 1111 for the stated fault target; retain 0000 for SA1. The hypothetical three-bit course encoding is used here, even though it is not a universal device opcode assignment.

All three instruction registers have the stipulated length three. Since the farthest IC's instruction is shifted first and each instruction is LSB-first, load TDI bits

$$\underbrace{000}_{IC3:\ Bypass}\quad\underbrace{010}_{IC2:\ ExTest}\quad\underbrace{010}_{IC1:\ ExTest}\ .$$

From Run-Test/Idle, the TMS sequence 1100 reaches Shift-IR. During the nine instruction shifts use eight zeros and a final 1, then 1,0 to reach Update-IR and return to Idle. From Idle, 100 reaches Shift-DR via Capture-DR. For a data register chain of length L, shift $L-1$ clocks with TMS=0 and the last with TMS=1; then use 1,0 for Update-DR and Idle. For response capture, take Idle→Select-DR→Capture-DR and then Shift-DR.

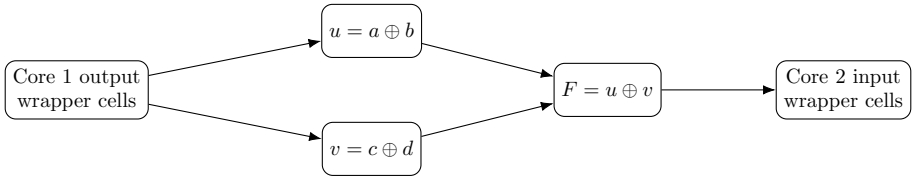
The figure does not specify serial ordering of the individual boundary cells, their enable values or a BSDL description. Therefore the full TDI data word cannot be uniquely inferred: it must encode 1111 at the four IC1 launch positions, safe values elsewhere, and one IC3 bypass bit. If each depicted IC has exactly eight boundary cells, $L = 8 + 8 + 1 = 17$; use the actual BSDL lengths/order in hardware. No assumption of arbitrary serial ordering should be hidden in a 17-bit answer.

Reset without TRST. Hold TMS=1 for at least five rising TCK edges, reaching Test-Logic-Reset from any TAP state. A subsequent TMS=0 enters Run-Test/Idle.

1149.1 versus 1500. IEEE 1149.1 primarily gives chip/board boundary access through a TAP controller and a standardized serial state machine. IEEE 1500 wraps embedded cores inside an SoC; it separates the wrapper from the system-level test access mechanism and supports wrapper serial access with optional parallel access for bandwidth. Thus their intended integration level and access/control architecture differ. See the TI boundary-scan design note and IEEE 1500 standard overview linked in `exams/README.md`.

6 Two wrapped cores and a three-bit TAM

Question. Two IEEE 1500-wrapped cores are connected by a three-bit daisy-chain TAM. The user-defined logic (UDL) is two two-input XORs feeding a third XOR. Instructions are Bypass, InTest and ExTest. (a) Generate tests for all UDL stuck-at faults. (b) Give the serial external-test data path and procedure. (c) Organize internal scan chains of lengths 120, 240, 240 and 120 to minimize scan time, with the same operating frequency and supply for both cores.



Solution. (a) Use $(a, b, c, d) \in \{0000, 1000, 0010, 1111\}$. Every primary input takes both values; each first-level XOR takes both values; and the last XOR takes both values. Any single stuck-at error at a stem or gate-input branch propagates through XORs without a controlling-value mask. Thus all such UDL faults are detected.

(b) Load ExTest in both wrapper instruction registers. Select each WBR rather than the core's internal scan chains or bypass register. The serial path is WSI→Core-1 WBR→Core-2 WBR→WSO, through the appropriate wrapper output selection muxes. Shift launch values, update Core-1 output cells, allow the UDL to settle, capture Core-2 input cells, then shift out and compare responses while

loading the next vector. The shared WCS supplies wrapper capture/shift/update controls; parallel TAM width does not by itself make this serial ExTest operation parallel.

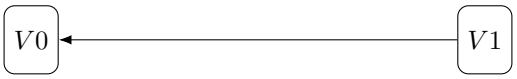
(c) Use the two directly connected parallel internal-scan lanes: chain 1 feeds chain 3, and chain 2 feeds chain 4. The path lengths are $120+240=360$ and $240+120=360$. Thus 360 shift clocks are sufficient for the internal-state data per vector. For P patterns and one capture clock per pattern, overlapping scan-out with the next scan-in gives $360(P+1)+P$ clocks, excluding control setup.

This follows the drawn mux connectivity: Core-1 mux c selects the upper TAM lane or chain-2 tail for chain 1; mux b selects the middle TAM lane or input-WBR tail for chain 2. Output mux g selects a fixed three-bit group of chain-1 tail, chain-2 tail and WBR tail, or the bypass group. Core-2 muxes j, i, n have the same structure. Choose $c=0, b=1, g=1, j=0, i=1, n=1$ for the two parallel lanes. There is no crossbar connecting chain 1 to chain 4 while independently placing chains 2 and 3 on the other two lanes. The third lane can be serialized through wrapper cells and internal chains, but is not a third independent internal-scan lane; wrapper cells then add length. The free-assignment lower bound $\max(240, \lceil 720/3 \rceil) = 240$ is therefore unattainable for the displayed routing. Include any extra pipeline/bypass bits when timing a physical implementation.

7 Memory faults, March control and BIST

Question. (a) Give an FSM for an idempotent coupling fault whose aggressor undergoes a down transition and forces the victim to 0. (b) Give an FSM implementing March C− and describe a pattern generator. (c) Name three basic memory-BIST components. (d) Additionally, for a DRAM with a 32 ms refresh period, derive a test for stuck-at and data-retention faults.

Solution. (a) Represent the victim by states $V0, V1$ and take a write-induced aggressor transition as the event. On the specified aggressor transition, force the victim state to 0; for all other nonvictim events, retain the victim. Ordinary writes to the victim set the corresponding state.



The arrow above denotes the erroneous forcing event. To detect it, initialize the victim to 1, initialize the aggressor to 1, perform the aggressor transition and read the victim before any legitimate write overwrites the corruption. The fault-free FSM leaves the victim unchanged.

(b) March C− is

$$\{\Updownarrow(w0); \Uparrow(r0, w1); \Uparrow(r1, w0); \Downarrow(r0, w1); \Downarrow(r1, w0); \Updownarrow(r0)\}.$$

Use the six controller states below; two-operation states have read and write subphases at each address.

State	Direction	Per-address operations	Next
S_0	either	$w0$	S_1
S_1	up	$r0, w1$	S_2
S_2	up	$r1, w0$	S_3
S_3	down	$r0, w1$	S_4
S_4	down	$r1, w0$	S_5
S_5	either	$r0$	Done

Advance the address only after all operations of the current element finish. At the last address, reload the appropriate end address and move to the next state. Generate write data and expected

read data from the state/subphase. The total is $10N$ memory operations for N addresses, ignoring setup cycles.

(c) A *pattern/address generator* emits addresses, data and read/write controls; a *controller* sequences the March elements and handles timing; a *response analyzer* compares read data with expected data and records pass/fail or failing addresses/bits. In a practical implementation, address and data generation may be separate datapath blocks under that controller.

(d) With refresh disabled during the intentional hold interval, write every cell to 0; hold for the specified retention interval (32 ms, with the chosen test margin); read every cell expecting 0. Then write every cell to 1, hold again and read expecting 1. This tests both stored polarities and exposes stuck values as well as loss of either stored value. Time each cell's hold from its completed write and ensure no refresh or intervening access restores it. Immediate March reads can test SAFs/transitions but cannot establish retention over 32 ms. Include temperature, voltage and refresh policy in an electrical test specification.